CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**Skills in Civil Engineering software. (STAADPRO/CAD/TEKLA)**	L	T	P	C
23A01706		0	1	2	2

Course Objectives

The objectives of this course are to:

1. Provide fundamental knowledge of **Auto CAD** and **Tekla** for 2D drafting and 3D modeling.
2. Train students in **structural analysis and design** using **STAAD.Pro**
3. Develop skills in **reinforcement detailing and structural component modeling**.
4. Introduce **seismic and nonlinear analysis** for structural safety evaluation.
5. Enable students to apply **civil engineering software tools** for real-world projects.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

1. Create **2D drawings and structural detailing** using **AutoCAD**.
2. Develop **3D building models** and analyze site to **pography** in **Revit**.
3. Perform **structural analysis and design** of **multi-story buildings** using **STAAD.Pro and ETABS**.
4. Conduct **seismic and performance-based analysis** for high-rise structures.
5. Apply **SAP2000** for**bridge modeling, water tank design, and advanced structural systems**.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	**3**	**1**	**1**	**-**	**2**	**-**	**-**	**1**	**-**	**1**	**1**	**1**	**2**	**1**
CO -2	**3**	**1**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**2**	**2**	**2**	**1**	**1**
CO -3	**3**	**1**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**2**	**2**	**2**	**2**	**2**
CO -4	**3**	**1**	**1**	**-**	**3**	**-**	**-**	**1**	**-**	**1**	**1**	**1**	**1**	**2**
CO -5	**3**	**1**	**1**	**-**	**3**	**-**	**-**	**1**	**-**	**1**	**1**	**1**	**1**	**1**

List of Experiments:-

1. Determination of Basic Drawing and Editing Commands in AutoCAD
2. Creation of a 2D Floor Plan for a Residential Building in AutoCAD
3. Development of Structural Detailing for Beams and Columns in AutoCAD
4. Application of Reinforcement Detailing for Slabs and Footings in AutoCAD
5. Creation and analysis of a steel column using TEKLA
6. Modelling a Roof Truss by TEKLA
7. Design of a Composite Beam by TEKLA
8. Column Base Plate Design by TEKLA
9. Determination of STAAD.Pro Interface and Structural Model Setup
10. Analysis and Design of a Simply Supported Beam in STAAD.Pro
11. Development of Structural Analysis for a Multi-Story RCC Building in STAAD.Pro
12. Application of Seismic Load Analysis on a Building Structure in STAAD.Pro

TEXT BOOKS:

1. **George Omura, Brian C. Benton – *Mastering AutoCAD 2025 and AutoCAD LT 2025*, Wiley, 2025 Edition**

2. **TEKLA Structural Designer 2023 Engineers Hand Book by Trimble Solutions Corporation**

REFRENCE BOOKS:

1. **Phil Read, Eddy Krygiel, James Vandezande – *BIM Handbook: A Guide to Building Information Modeling for Owners, Designers, Engineers, and Contractors*, John Wiley & Sons, 4th Edition, 2023**
2. **NighatYasmin Ph.D., *Introduction to AutoCAD 2025 for Civil Engineering Applications*, SDC Publications, 2024 Edition.**